INCH-POUND
MIL-DTL-83725/16C
20 January 2022
SUPERSEDING
MIL-R-83725/16B(USAF)
28 September 1987

DETAIL SPECIFICATION SHEET

RELAYS, VACUUM, SPDT, 12 AMPERES DC OR 60 HZ RMS, 9 KILOVOLTS DC OR 60 HZ PEAK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-DTL-83725.

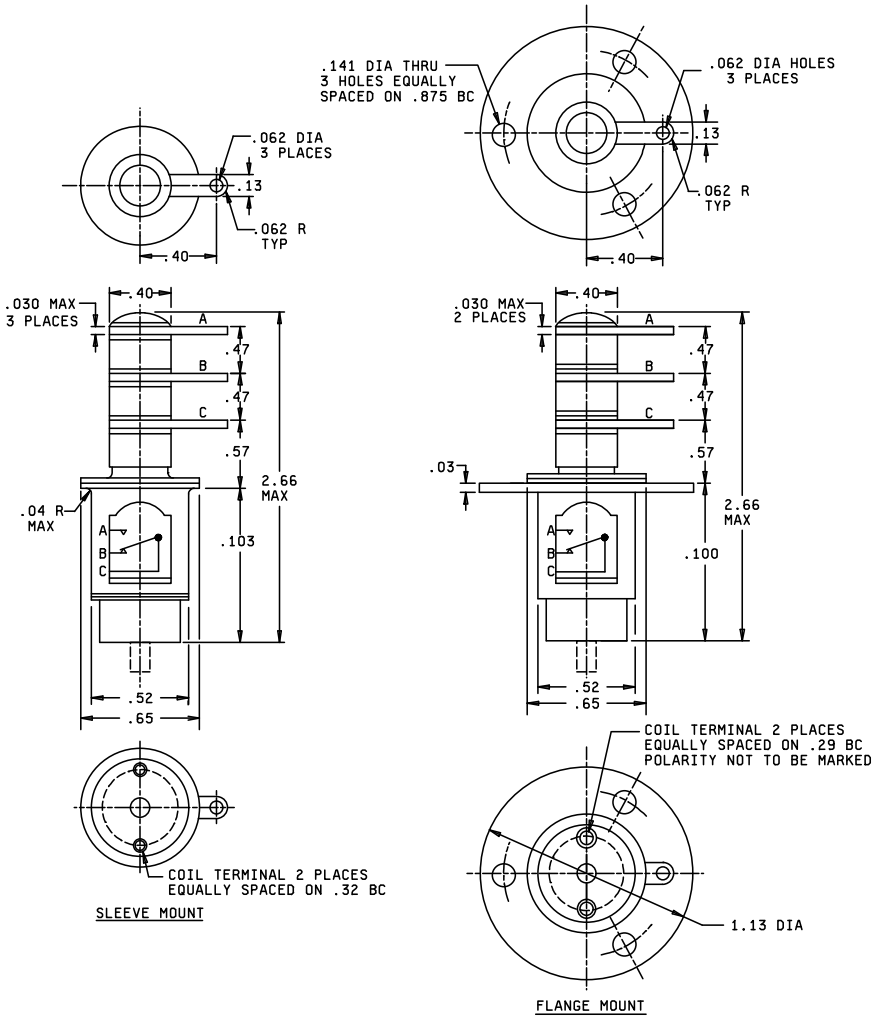
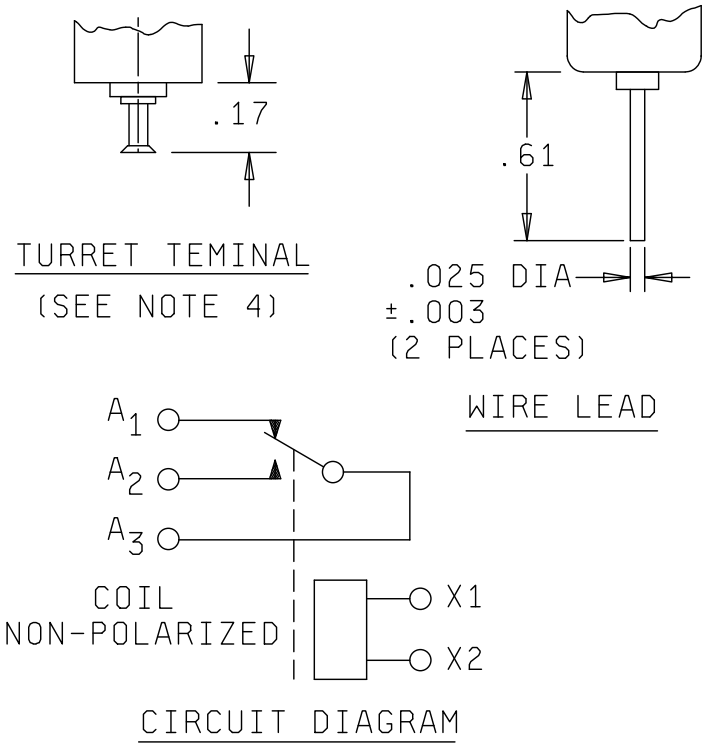


FIGURE 1. Relay, dimensions, and configurations.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:26Z
Check the source to verify that this is the current version before use.



Inches	mm	Inches	mm	Inches	mm
.003	0.08	.17	4.3	.57	14.5
.015	0.38	.32	8.1	.61	15.5
.025	0.64	.40	10.2	.65	16.5
.030	0.76	.47	11.9	.875	22.22
.04	1.0	.52	13.2	1.00	25.4
.062	1.57	.54	13.7	1.03	26.2
.13	3.3			1.13	28.7
.141	3.58			2.59	65.8
				2.66	67.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.5 mm) for two place decimals.
4. Shape of coil lug (solder) terminals optional, however they must accommodate two no. 22 AWG wires.
5. Glazing of ceramic insulator is option.
6. MS83725/16-001, -002, -005, and -006 mounted by .52 inch (13.2 mm) diameter of body of coil housing or by .65 inch (16.5 mm) diameter lip in center of relay.

FIGURE 1. Relay, dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:26Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

CONTACT DATA:

Configuration: SPDT, ground isolated.

Load ratings:

Resistive: (Carry only.)

12 amperes dc or 60 Hz rms.

10 amperes rms, 2.5 MHz.

6 amperes rms, 16 MHz.

Rated operating voltage:

9 kilovolts peak, 60 Hz or dc.

7 kilovolts peak, 2.5 MHz.

6 kilovolts peak, 16 MHz.

Contact resistance:

Rated Life:

Before: .020 ohm, maximum.

During: .100 ohm, maximum.

After: .100 ohm, maximum.

Capacitance:

1.6 picofarads, maximum across open contacts.

1.6 picofarads, maximum between open contacts and ground.

COIL DATA: (See table I.)

Duty rating: Continuous.

Operate time: 10 milliseconds with nominal coil voltage.

Release time: 10 milliseconds from nominal coil voltage.

TABLE I. Operating characteristics and dash numbers

Dash number M83725/16-	Coil data		At 25°C				Over temp. range			Mounting	Terminal type
	Nominal (V dc) 1/	Maximum (V dc)	Coil resis-tance ohms	Pickup voltage (V dc)	Hold voltage (V dc)	Dropout voltage (V dc)	Pickup voltage (V dc)	Hold voltage (V dc)	Dropout voltage (V dc)		
-001	12	16	43 to 77	8.0	5.0	0.5	11	7.0	0.3	Sleeve	Turret
-002	26.5	32	290±10%	16	10	1.0	22	14	0.7	"	"
-003	12	16	43 to 77	8.0	5.0	0.5	11	7.0	0.3	Flange	"
-004	26.5	32	290±10%	16	10	1.0	22	14	0.7	"	"
-005	12	16	43 to 77	8.0	5.0	0.5	11	7.0	0.3	Sleeve	Wirelead
-006	26.5	32	290±10%	16	10	1.0	22	14	0.7	"	"
-007	12	16	43 to 77	8.0	5.0	0.5	11	7.0	0.3	Flange	"
-008	26.5	32	290±10%	16	10	1.0	22	14	0.7	"	"

1/ CAUTION: Use of any coil voltage less than nominal voltage will compromise the operation of the relay.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:26Z
Check the source to verify that this is the current version before use.

ELECTRICAL DATA:

Insulation resistance: 1,000 megohms, minimum, except the resistance between coil and housing at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage:

At atmospheric pressure: 60 Hz.

Between all mated contacts in the open position: 9 kV peak.

Between high voltage terminals and housing: 9 kV peak.

Between coil and housing: 500 V rms, 60 Hz.

Ground isolated: Ground isolated relays have operating elements insulated from the ground plane to provide absolute voltage isolation between the frame and the high voltage contacts.

ENVIRONMENTAL DATA:

Temperature range: -55°C to +125°C.

Vibration: MIL-STD-202-204, test condition C (10 to 55 Hz, .06-inch DA/55-2000 Hz, 10 g's)

Shock: MIL-STD-202- 213, test condition J, 30 g's. 11 milliseconds, half sine.

PHYSICAL DATA:

Terminal strength: 5 pounds pull.

Dimensions and configuration: See figure 1.

Termination: Solder terminal.

Weight: 1 ounce, maximum.

LIFE TEST REQUIREMENTS:

Mechanical cycling: 1,000,000 cycles. Two sample units (cycling rate, 36,000 per hour maximum; contact current shall not exceed 10 milliamperes).

QUALITY ASSURANCE:

Dielectric withstanding voltage:

Tests to be conducted at atmospheric pressure rating only.

Duration of application: 5 to 10 seconds at a 10-percent increase in the dielectric withstanding voltage.

PART NUMBER: M83725/16- (dash number from table I).

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-DTL-83725, this document references the following:

MIL-STD-202-204 MIL-STD-202--213

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:26Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy – AS
Air Force – 85

Review activity:
Navy – OS

Preparing activity:
DLA-CC

(Project 5945-2022-012)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:26Z
Check the source to verify that this is the current version before use.